GG-S-00945d (VAMI)
May 21, 1969
SUPERSEDING
INT. FED. SPEC.
GG-S-00945c (VAMI)
NOVEMBER 15, 1965

INTERIM FEDERAL SPECIFICATION
SYRINGE AND HYPODERMIC NEEDLE, DISPOSABLE, STERILE

This Interim Federal Specification was developed by the Veterans Administration, Department of Medicine and Surgery, Supply Service, Marketing Division for Dental and Surgical Supplies, Marketing Center, P.O. Box 76, Hines, Illinois 60141, based upon currently available technical information. It is recommended that Federal Agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers sterile disposable plastic syringes and hypodermic needles suitable for one time use in medical procedures.

1.2 Classification. Syringes and needles shall be of the following types, styles, classes, sizes and gages as specified in the invitation for bids or requests for proposals: (See 6.1).

1.2.1 Types, Styles and Classes

Type I Plunger Type

Type II Plungerless Type

Type III Plunger type with Finger Control (Control Syringe)

Style A Without Needle

Style B With Needle

FSC 6515

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Class 1 Luer Tip

Class 2 Catheter Tip

Class 3 Tapered Tip

Class 4 Luer-Lock Tip

1.2.2 Size, gages, point styles

1.2.2.1 Syringe Sizes

1 cc Insulin

1 cc Insulin short model

1 cc Tuberculin

2½ - 3 cc General Purpose

5 cc to 6 cc General Purpose

10 cc to 12 cc General Purpose

20 cc General Purpose

30 cc to 35 cc General Purpose

50 cc to 60 cc General Purpose

2oz/50 to 60 cc Aspirating/Irrigating

3oz/100 cc Aspirating/Irrigating

2-3oz Aspirating/Irrigating

1.2.2.2 Needle Gages

18 gage	23 gage
19 gage (thinwall)	25 gage
20 gage	26 gage
21 gage	27 gage
22 gage	

2.

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1.2.2.2.1 Point Styles

Style A - Regular bevel (for subcutaneous and intramuscular injection)

Style B - Short bevel (for intravenous injection)

Style C - Very short bevel (for intradermal injection)

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specifications and standards of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification.

FEDERAL SPECIFICATION:

PPP-B-566c Boxes, Folding Paperboard

PPP-B-585b Boxes, Wood, Wirebound

PPP-B-601c Boxes, Wood, Cleated-Plywood

PPP-B-621b Boxes, Wood, Nailed and lock-corner

PPP-B-636c Boxes, Fiberboard

PPP-B-676c Boxes, Set-up

FEDERAL STANDARDS:

Federal Standard No. 102 - Preservation, Packaging, and Packing Levels

Federal Standard No. 123 - Marking for Domestic Shipment (Civilian Agencies).

FOR INSTRUCTION ON OBTAINING COPIES OF FEDERAL SPECIFICATIONS AND STANDARDS, SEE LAST PAGE.

MILITARY SPECIFICATIONS

MIL-L-10547 Liner, Case, and Sheet, Overwrap, Water-Vaporproof or Waterproof, Flexible

MIL-L-36124 Needle, Hypodermic, Disposable

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MILITARY STANDARDS

MIL-STD-105d Sampling Procedures and Tables for Inspection by Attributes

MIL-STD-129d Marking for Shipment and Storage

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other Publications. The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids or requests for proposals shall apply.

AMERICAN STANDARDS ASSOCIATION, INCORPORATED: (ASA)

ASA-70.1 - 1955 (UDC615.47) American Standard Dimensions of Glass and Metal Luer Tapers for Medical Applications.

(Application for copies should be addressed to the American Standards Association, Incorporated, 10 East Fortieth Street, New York, New York (10016).

AMERICAN IRON AND STEEL INSTITUTE (AISI)

Steel Products Manual, Stainless and Heat-Resisting Steels.

(Application for copies should be addressed to the American Iron and Steel Institute, 150 East Forty-Second Street, New York, New York (10017).

UNITED STATES PHARMACOPOEIAL CONVENTION PUBLICATION: (USP)

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Pharmacopoeial of the United States.

(Application for copies should be addressed to the Mack Printing Company, Easton, Pennsylvania (10843).

FEDERAL FOOD, DRUG AND COSMETIC ACT AND GENERAL REGULATIONS FOR ITS ENFORCEMENT

(Copies are available from Superintendent of Documents, Government Printing Office, Washington, D.C. (20420).

3. REQUIREMENTS

3.1 Material. All material shall comply with the requirements of the Federal Food, Drug and Cosmetic Act. The portions of the unit that come in contact with the fluid to be injected shall be compatible with all known medications or if not, each individual package shall have a warning or caution. (See 5.3).

3.1.1 Plastic. Plastic shall be non-toxic, high grade, and shall not have more than 5 p.p.m. extractable heavy-metals (see 4.3.3). Plastic shall be polypropylene or polystyrene as specified, or other material meeting the requirements of 3.1 and suitable in all respects for the purpose intended.

3.1.2 Rubber. Rubber shall be fresh stock, pharmaceutical grade, non-blooming, non-leaching, non-toxic, age-resistant, inert, natural or synthetic or combinations thereof processed for the specific purpose (see 4.3.3).

3.1.3 Lubricant (Coating). The lubricant shall be an inert, pharmaceutical grade non-toxic, silicone compound or

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equal, suitable for the specified use. It shall serve as a coating on all interior surfaces of the syringe that come in contact with the plunger tip. The distal end of the needle cannula shall be coated.

3.1.4 Metal. Metal parts, other than needle cannula, shall be fabricated of suitable aluminum alloy degreased and anodized or degreased and passivated, stainless steel, nickel plated brass or nickel plated copper, suitable for the purpose intended.

3.1.4.1 Steel Tubing, Steel tubing shall be either seamless drawn or welded and drawn, (AISI), Type 304 corrosion resistant steel and withstand the applicable test specified in Section 4.

3.1.5 Cotton. When cotton is used as a protective filter on open-tip type needle protectors, it shall be strict middling grade, bleached cotton of staple length 1-1/16 inch to 1-1/8 inch.

3.1.6 Adhesive. Adhesive if used to connect the cannula to the syringe tip or cannula to hub, shall be inert, non-toxic (see 3.1) and withstand the applicable tests in section 4.

3.2 Design and Dimensions. Type I syringes shall consist of a barrel, plunger, and needle or tip protector. Style A shall be supplied without a needle and Style B with a needle of the gage, length, and point style specified. The needle shall be

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securely affixed on or in the barrel tip, ready for use, and shall be sufficiently set to prevent its dislodgment during shipment, handling and normal use. Type II shall be an aspirating, irrigating syringe, consisting of a barrel and bulb or bellows and with or without a tip protector. Type III shall be a plunger type syringe as described above but additionally equipped with removable or attached finger and thumb grip controls.

3.2.1 Syringe, Type 1.

3.2.1.1 Barrel. The barrel shall be a rigid tube fabricated, unless otherwise specified, (see 6.1), of polypropylene (see 3.1.1), closed at one end and except for an orifice and open at the other end to receive the plunger. At the closed end there shall be formed a tip. The wall at the open end shall be thickened and extend radially outward to form a flange to prevent slipping of the barrel through the fingers, opposite segments of the flange shall provide straight parallel sides of ample length to prevent rolling. The inside of the barrel shall have a very fine, uniform surface, free from pits, bubbles, mold marks, open air lines or other defects that would affect the fluid tightness and smoothness of its fit with the plunger. The inside of the barrel shall be lightly coated with the lubricant specified in 3.1.3. Inside diameter shall be such as to insure an accurate fit with the plunger tip and result in a fluid and air tight fit (see 4.3.1), and yet permit plunger to move freely through out its excursion in the barrel (see 4.3.1.3). Ungraduated portion of the barrel,

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at the proximal end, shall be of sufficient length to permit plunger tip to align with the top graduation mark while the barrel is encompassing the entire plunger tip, but shall be not greater than one-third the overall length of the syringe barrel. Dimensions of the barrel shall be as specified in Table I.

3.2.1.1.1 Tips. Tips, unless otherwise specified (see 6.1), shall be formed concentric to the axis of the barrel. Orifice through the tip shall be concentric therewith and of diameter specified (see Table I).

3.2.1.1.1.1 Luer tip. Luer tip shall make a mechanically stable, fluid and air tight connection with a hypodermic needle having a luer hub with a hub cavity in accordance with figure 2 (female end) of ASA standard Z-70.1-1955.

3.2.1.1.1.2 Catheter Tip. Catheter tip shall measure 1.450 $\pm$ .30 inches overall length, and uniformly taper from 0.4050 $\pm$ 0.0925 inches to 0.250 $\pm$ 0.0312 inches diameter and have an effluent bore of 0.172 $\pm$ 0.04 inches.

3.2.1.1.1.3 Taper Tip. The taper of the tip shall be gradual and consistent and shall have a constricted area at the distal end approximately 0.50 inch in length, with the walls practically parallel. Tip shall have a diameter of 0.250 $\pm$ 0.0312 inches at the distal end and have an effluent bore of 0.141 $\pm$ 0.024 inches.

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TABLE I

CAPACITY ±-5%	LENGTH OVERALL INCLUDING TIP WITHOUT NEEDLE	DIAMETER OUTSIDE (NEAR DISTAL END)	DIAMETER INSIDE	DIAMETER ORIFICE	GRADUATIONS SUB UNLESS OTHERWISE SPECIFIED	GRADUATIONS MAIN
CC/OZ	INCHES	INCHES	INCHES	INCHES	CC	CC
1 cc short	2.10-2.60	0.25-0.35	0.20-0.30	.015-.086	SEE 3.2.1.1.2	SEE 3.2.1.1.2
1 cc long	3.130-4.00	0.20-0.30	0.16-0.20	.015-.086	SEE 3.2.1.1.2	SEE 3.2.1.1.2
2½-3cc	2.10-3.10	0.40-0.50	0.30-0.42	.030-.086	0.2	1.0
5-6 cc	2.75-3.50	0.45-0.75	0.40-0.60	.030-.086	0.2	1.0
10-12 cc	3.50-4.30	0.60-0.85	0.55-0.70	.030-.086	1.0	5.0
20 cc	3.75-4.75	0.75-1.00	0.62-0.88	.030-.086	1.0	5.0
30-35 cc	3.90-5.00	0.80-1.45	0.62-1.00	.030-.086	5.0	10.0
50-60 cc	5.00-6.25	1.10-1.62	0.70-1.25	.030-.086	SEE 3.2.1.1.1.3	SEE 3.2.1.1.2
50 cc 2 oz/60 cc	6.25-9.00	1.10-1.375	1.00-1.25	3.2.1.1.1.2	3.2.1.1.2.4	3.2.2.1.1
2-3oz	7.25-9.25	0.68-1.20			3.2.2.1.1	

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3.2.1.1.1.4 Luer lock tip. Tip shall have a standard luer taper of .060/in. per inch as defined in ASA Standard Z-70.1 - 1955 and be provided with a locking collar having an internal thread designed to accommodate the locking head on a plastic or metal needle hub.

3.2.1.1.2 Graduations and Markings. The graduation scale specified (see Table I) shall be applied to the outside surface of the barrel, in such a manner as to be approximately in line with the flattened side of the flange. The "0" mark of the scale shall register precisely with the distal ring of the plunger tip when the plunger is pushed to the bottom of the barrel. Main graduation marks shall be at least 33-1/3 percent longer than sub-graduation marks. The ungraduated portion shall be not greater than one-third the overall length of the syringe. No number is required at the "0" mark. In addition the outside surface of the barrel shall have marked thereon the name or registered trademark of the manufacturer, the words, "Discard After Use" or other similar legend denoting one time use. As an alternate, the legend denoting one time use may be on the individual package. Graduations and markings shall be legible and easily read. The graduations and marking shall not be removed, smeared, or faded to the extent of being illegible during use or shipment.

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3.2.1.1.2.1 1cc Insulin Syringe. Insulin syringe, 40/80 unit unless otherwise specified (see 6.1) shall have a double scale. The 40 unit scale shall have main graduations at 5 - unit intervals with the alternate graduations marked 10, 20, 30 and 40, with subgraduations at 1-unit intervals unmarked. The 80 unit scale shall have main graduation at 5-unit intervals marked 10, 20, 30, 40, 50, 60, 70, and 80 with subgraduations at 1-unit intervals unmarked. Scale for 40 unit syringe shall be marked in a shade of red or reddish brown. The scale on the 80 unit shall be marked in a shade of green. Each scale shall bear the words, "UNITS", and the capacity clearly marked. The same interval graduations, markings and colors shall apply on 40 unit or 80 unit single scale insulin syringes.

3.2.1.1.2.2 1cc Tuberculin Syringe. Tuberculin syringe shall have a double scale, one in cubic centimeters with main graduations at 0.20cc., intervals marked 20, 40, 60, 80 and 1cc., successively, with subgraduations at 0.01 or 0.02cc., intervals unmarked. As an alternate, the cc., scale shall have main graduations at 0.10cc., intervals marked 10, 20, 30, 40, 50, 60, 70, 80, 90 & 1cc successively, with subgraduations at 0.01 or 0.02cc intervals unmarked. The other scale shall be a minim scale with graduations at 4 minim intervals marked 4, 8, 12 and 16 successively, with subgraduations at 0.5 minim intervals unmarked.

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3.2.1.1.2.3 2½-3cc General Purpose. Syringe shall have a double scale. One in cubic centimeters with main graduations at 0.5cc. intervals marked not less than ½, 1, ½, 2 and 2½cc. successively with subgraduations at 0.1cc. unmarked. The other shall be a minim scale with main graduations at 5 minim intervals marked to show not less than 30 minim graduation with subgraduations at 1 minim intervals unmarked.

3.2.1.1.2.4 2oz/50 to 60cc. Syringe shall have a double scale. One in cubic centimeters with main graduations at 10cc intervals, marked 10, 20, 30, 40, 50 and 60 cc (if applicable) successively with subgraduations at 5cc. intervals unmarked. As an alternate the subgraduations may be omitted. The other shall be in ounces with main graduations at ½ ounces marked ½, 1, ½, and 2oz., successively, with subgraduations at ½ ounce intervals unmarked. As an alternate main graduations shall be at 1 oz. intervals marked 1 and 2 oz. successively and without subgraduations.

3.2.1.2 Plunger. Plunger shall be fabricated of the material specified in 3.1.1. The distal end of the plunger shall be shaped to permit secure attachment of a plunger tip, as an alternate the plunger shall be shaped to form the tip as an integral part of the plunger and of the same material. The proximal end of the plunger shall extend well out of the barrel,

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and shall be shaped to form a knob of a thickness and diameter appropriate to the size of the syringe except when formed to one design of a control syringe, (see 3.2.1.3). The distal end of each plunger with its attached or molded plunger tip shall accurately fit its individual barrel, permit the plunger and plunger tip to draw and expel fluids, and yet move freely and smoothly throughout its excursion of the barrel. The plunger tip of material specified in 3.1.2 shall be securely attached to the plunger so as not to become loose or detached in use. The tip shall be shaped to closely fit the barrel and shall be so designed as to permit easy and efficient removal of air bubbles. The plunger tip shall be properly treated with the lubricant that detracts from appearance, interferes with visibility of the plunger tip ring, or can be washed out with the solution used, shall be cause for rejection, (see 4.3.6). Graduation markings, denoting correct, accurate capacity shall register precisely with the first ring at the plunger tip distal end, touching sides of the barrel. Tip shall be of permanent suitable color so that the first ring at the plunger tip distal end, touching sides of the barrel, shall be clearly visible and shall accurately indicate through the barrel wall. Plunger and tip shall withstand applicable tests in Section 4.

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3.2.1.3 Plunger design and accessories for Type III, Control Syringe. The plunger for the control syringe shall have either the knob described in 3.2.1.2 or shall have as an integral part of the plunger a thumb ring of adequate size and shape for maximum efficiency and comfort. If a knob is furnished, the thumb ring may be separately attached but shall fit tightly and securely. The Type III syringes shall also have either as an integral part of the barrel or as separate but securely attached to the barrel two flanges or finger rings of sufficient size and shape for maximum efficiency and comfort.

3.2.2 Syringe, Type II. The Type II syringe shall be fabricated of plastic (see 3.1.1). The barrel shall consist of a tube closed at the distal and except for a tip with orifice of diameter specified (see Table I) and having a flexible cylindrical bulb or bellows at the proximal end. The bulb or bellows shall be integrally molded or secured to the barrel in such a manner as to insure against leakage during suction and compression. Bulb or bellows shall have a capacity of not less than 1/5 more than the maximum graduation shown and shall completely fill to the maximum graduation and/or empty with one steady compression.

3.2.2.1 Graduations and Other Markings. The outside surface of the barrel shall have marked thereon the name or

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registered trademark of the manufacturer, as an alternate the name or registered trademark may be on the individual package. The outside surface of the bulb, bellows or barrel shall have marked thereon the scale specified (see 3.2.2.1.1). Markings shall be easily read, either molded or marked. If marked they shall remain legible during use and shipment.

3.2.2.1.1 2oz/3oz. Syringe shall have a double scale, one marked in cubic centimeter at 10cc main graduations, the other marked with main graduations in ounces.

3.2.3 Needle. The needle shall be of the gage, length, and point style, if other than regular, specified (see 6.1) and shall consist of a cannula and hub.

3.2.3.1 Cannula. The cannula shall be a straight tube of material specified in 3.1.4.1 and, shall be lubricated with material specified in 3.1.3. The bevel shall be ground concave or flat and may have a center ridge at the distal end to strengthen the tip, shall be so ground with a side bevel at the outer end, except point style C may be ground without side bevel, as to enhance the keenness of the tip and cutting edges. Both inner and outer edges of the point shall be free of burrs and slivers. The tip of the point shall be straight and strong and of maximum keenness. The proximal end of the cannula shall extend well into the hub or syringe tip sufficiently to allow for secure affixing. The cannula shall be centered in such

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a manner as to provide a straight, concentric, continuous fluid path from the syringe tip. The cannula shall not deviate more than 3 degrees at needle point, from the longitudinal axis of the syringe tip. Cannula shall withstand the applicable tests specified in Section 4. Dimensions of the cannula, other than length, shall be as indicated in Table II for the appropriate gage. The cannula length shall be measured from the tip of the point to the junction with the hub or syringe tip. Tolerance for cannula length shall be as indicated in Table III.

TABLE II
DIMENSIONS OF NEEDLES

GAGE	CANNULA DIAMETER LIMITS (INCH) OUTSIDE MINIMUM	OUTSIDE MAXIMUM	MINIMUM INSIDE	STYLE	LENGTH (INCH) MAXIMUM	LENGTH (INCH) MINIMUM
27	.0155	.0165	.0075	a	.085	.070
				c	.052	.042
26	.0175	.0185	.0093	a	.095	.072
				c	.052	.042
25	.0195	.0205	.0095	a	.103	.085
				b	.077	.063
23	.0245	.0255	.0125	a	.129	.105
				b	.086	.070
22	.0275	.0285	.0155	a	.146	.120
				b	.095	.077
21	.0315	.0325	.0195	a	.163	.133
				b	.104	.084
20	.0340	.0360	.0220	a	.180	.148
				b	.120	.093
18	.0475	.0505	.0315	a	.249	.203
				b	.163	.133
19 TW	.0405	.0435	.0300	a	.215	.176
				b	.138	.113

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TABLE III
TOLERANCES FOR CANNULA LENGTH

LENGTH (INCH)	TOLERANCES (INCH)
To 1/2	± 1/32
1/2 to 3/4	± 3/64
3/4 thru 1½	± 1/16
1-3/4 thru 2½	± 5/64
3 and longer	± 1/8

3.2.3.2 Hub. Hub, unless otherwise specified (see 6.1) shall be fabricated of material specified in 3.1.1 or 3.1.4. If hub is made of brass the entire hub cavity wall, down to the entrance of the cannula, as well as the exterior of the hub shall be well plated with chrome or nickel. Similarly if fabricated of aluminum, the anodization or passivation shall include the entire hub cavity, down to the entrance of the cannula, as well as the exterior of the hub. A metal insert may be used for connecting the hub to the cannula. If insert is fabricated of brass or copper alloy (see 3.1.4) the entire insert cavity wall down to the entrance of the cannula, as well as the exterior of the insert, shall be well plated with chrome or nickel. The hub and hub cavity shall be in accordance with figure 3 of ASA standard Z70.1-1955, except, the hub shall be luer slip or luer lock unless

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otherwise specified (see 6.1). A slight modification to the taper length of the hub shall be permitted provided such modification will insure a fluid and air tight fit. Hubs shall withstand the applicable tests in section 4.

3.2.4 Protector, Needle and Syringe Tip. Protector shall be fabricated of rigid material (see 3.1.1), suitable for the purpose intended, so designed as to fit securely in place during shipment and normal handling. The protector shall be so designed as to prevent it from shattering when used as a means of breaking the tip of the syringe to render it unusable.

3.3 Accuracy. When tested as specified in 4.3.1.4, syringes shall deliver plus or minus 5 percent of the capacity indicated at each graduation mark tested.

3.4 Freedom from Foreign Material. When tested as specified in 4.3.6 the fluid path of the syringes and needles as specified, shall be free from chips, lint, dirt, or other foreign material.

3.5 Safety (Toxicity) and Lead Content.

3.5.1 Toxicity. When tested as specified in 4.3.3.1 the syringe and needle, Type I and III, shall comply with the Safety test for Transfusion and Infusion Assemblies of the current edition of the U.S.P.

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3.5.2 Lead Content. When tested as specified in 4.3.3.2 all plastic and rubber components coming in contact with the medication being administered shall contain not more than 5 parts per million (p.p.m.) of lead.

3.6 Freedom from Pyrogens. When tested as specified in 4.3.4 the syringes and needles, Type I and III, shall comply with the pyrogen test requirements of the current edition of U.S.P.

3.7 Sterility. Syringes and needles shall be sterilized by any commercially acceptable procedure and shall comply with the requirements of the current edition of the U.S.P. when tested as specified in 4.3.5.

3.8 Certificates. Certificates, when requested (see 6.1) listing all lot control numbers, shall state that samples from each lot or control number have been tested as required in 3.5 and 4.3.3; 3.6 and 4.3.4; and 3.7 and 4.3.5; and have been found to comply with the test requirements. The certificate shall state the method used in the test.

3.9 Lot Control Numbers. Lot control numbers shall be clearly marked (see 5.3).

3.10 Instructions. The first multiple package, either unit of issue or if unit of issue is one, the intermediate

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and/or shipping container shall be provided with instructions describing the proper techniques for dispensing the sterile syringe and needle, as specified, from the individual container and preparing them for use. In addition information shall be shown as to the method for rendering the syringe and needle, as applicable, unusable.

3.11 Workmanship. Workmanship shall be first class throughout. Syringes and needles as applicable, shall be free from any defects affecting appearance, safety, serviceability, or for use for which intended.

4. SAMPLING, INSPECTION AND TEST PROCEDURES.

4.1 Responsibility for Inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that the supplies and services conform to prescribed requirements.

4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual, auditory, etc., investigation without the use of special laboratory appliances or procedures) and testing (determination by technical

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means of physical and chemical properties) of the item.

4.1.2 Certificates of Quality. When available, certificates of quality, supplied by the manufacturer of the component material, listing the specified test method and test results obtained, may be furnished in lieu of actual performance of such testing by the contractor.

4.2 Sampling.

4.2.1 Civilian. One-half of 1 percent of delivery, but not less than five samples, shall be taken at random from each 1,000 of the same size or fraction thereof, except when the shipment is over 10,000 of the same size, in which case at least two samples shall be taken from each one-tenth of the shipment, for examination for compliance with this specification.

4.2.2 Military. Sampling procedures shall be conducted in accordance with the procedures set forth in Military Standard MIL-STD 105 and unless otherwise specified (see 6.1) shall be based on the unit of product being one syringe or needle as specified.

4.2.2.1 Syringes.

4.2.2.1.1 Inspection lot. The inspection lot shall be determined in accordance with Military Standard MIL-STD 105 and, unless otherwise specified, shall be based on the unit of product being one syringe (end item).

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4.2.2.1.2 Sampling for examination. A random sample of syringes shall be selected from each lot in accordance with Military Standard MIL-STD 105. The inspection level and Acceptance Quality Level (AQL) shall be as indicated in Table IV.

TABLE IV

SAMPLING FOR EXAMINATION

Type Defect	Inspection level*	(AQL) (percent defective)
Major A	S-4	1.0
Major B	S-1	1.0
Minor A	S-4	6.5
Minor B	S-1	4.0

* If acceptance number is not exceeded, lot shall be accepted. If rejection number is exceeded, inspection at level II may be performed to determine rejection or acceptance.

4.3 Tests.

4.3.1 Syringes Type I and III.

4.3.1.1 Pressure Test. The syringe, with or without needle, as specified, shall be attached to a source of water pressure equipped with a gage. Water used as a testing fluid shall be let in through the syringe tip or needle, until the pressure specified in Table V is registered on the gage. The plunger shall be drawn to a point half the graduated length

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where it shall be allowed to remain for 10 seconds. The plunger shall then be drawn back to midway between the first position and the top graduation mark where it shall be allowed to remain for 10 seconds. The plunger shall then be drawn back to the top graduation where it shall be allowed to remain for 10 seconds. The syringe shall be tested in both horizontal and vertical positions. During each test, enough water shall not pass the connection of the cannula with the hub, the connection of the hub with the syringe tip, or the connection of the cannula with syringe tip where no hub is used, or plunger gasket, to produce a falling drop. After each successive test, any water that may have passed shall be wiped away and the next test applied. (Note: Prior to testing syringes with needles, the needle hub shall be secured to the syringe tip in accordance with manufacturer's instructions.)

4.3.1.2 Suction Test. Water, as specified in Table V, shall be drawn into syringe through a hypodermic needle. An attempt shall be made to remove any bubbles that may have formed. The needle end shall then be sealed and with the syringe held in a vertical position, needle pointing up, the plunger withdrawn to the top graduation mark and held in this position for 10 seconds. There shall be no leakage of air past the plunger tip, as indicated by bubbling in the water.

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(Note: Prior to testing syringes with needles, the needle hub shall be secured to the syringe tip in accordance with manufacturer's instructions.)

4.3.1.3 Plunger Movement Test. An adapter to fit both the source of hydrostatic pressure and the syringe tip and equipped with a gage shall be used. Ten samples, not subject to previous plunger movement shall be selected. Plungers shall be moved once from the packed position to full capacity and then return to the full closed position, then subjected to pressure until movement of the plunger is noted. Scale of gage shall be read and reading noted. Reading of ten tests shall be totaled and averaged. Average of the 10 readings shall not exceed the pressures specified in Table V.

TABLE V

	SUCTION TEST	PRESSURE TEST	PLUNGER MOVEMENT TEST
CAPACITY	VOLUME OF WATER CC	POUNDS PER SQUARE INCH	
1cc	0.25	35	50
2½ - 3cc	0.5	30	35
5 - 6cc	2.0	25	25
10 - 12cc	5.0	20	20
20cc	10.0	20	15
30 - 35cc	15.0	20	12
50 - 60cc	25.0	20	10
2oz/50cc	25.0	20	10

4.3.1.4 Accuracy. Type I and Type III syringes shall be tested for accuracy at each main graduation mark (see Table I)

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using water and a calibrated burette. As an alternate water shall be discharged into a dry flask having the same nominal capacity as the syringe. Failure to pass the test shall be cause for rejection.

4.3.2 Needle.

4.3.2.1 Security of Attachment. The barrel of the syringe shall be firmly held in suitable fixture with the tip and needle free and exposed. A tensile load of 5 pounds, shall be gradually applied to the needle and allowed to remain for one minute. The cannula shall not separate from the hub, from the syringe when no hub is used, or the hub from the syringe tip. As an alternate, if the manufacturer provides instructions for securing or setting the needle to the syringe tip, these instructions shall be followed before test is performed.

4.3.2.2 Stiffness Test. The cannula shall be placed across two test edges (0.002 inch radius maximum) separated by the distance indicated in Table VI for appropriate gage. A load, as indicated in Table VI shall be applied at the center of the span and the resulting maximum deflection of the cannula shall be measured using a gage with graduations of at least 0.001 inch. The force shall be applied through an edge with a maximum radius of 0.036 inch and centered to within $\pm$ 0.003 inch. The maximum deflection of the cannula shall not exceed

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the distance indicated in Table VI for the appropriate gage. Failure to pass shall be cause for rejection.

4.3.2.3 Resistance to Breakage. The needle hub shall be held in a fixed position and a force applied to the cannula as indicated in Table VI, in order to bend the cannula in a clockwise direction through an arc of 25 degrees from the original position. The operation shall then be reversed (i.e. bent 25 degrees from the neutral position in a counter-clockwise direction) and finally the cannula returned to the neutral position. This complete operation (or cycle) shall be repeated 20 times. As an alternate, for needles without hubs, the distal end of the cannula shall be held in a fixed position. Breakage resulting from this test shall be cause for rejection.

4.3.2.4 Corrosion-resistance of Cannula. The cannula shall be immersed in 10 percent citric acid at room temperature for 5 hours and after removal from the acid, boiled in distilled water for 30 minutes, followed by immersion in distilled water at room temperature for 48 hours. At the end of the 48 hour period, the cannula shall be allowed to dry by evaporation, and shall then be examined. Evidence of corrosion shall be cause for rejection.

4.3.2.5 Sharpness. Point shall be subjected to visual examination, under good illumination, with a magnifier of

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4 to 7 power. Presence of hooks, burrs, slivers, and lack of uniformity of grind and point sharpness shall be cause for rejection.

4.3.2.6 Lubricant Test. Cannula shall be pushed through a thin rubber diaphram. Evidence of the coating shall be apparent by the reduction in cannula drag when compared with an uncoated needle.

4.3.2.7 Luer Hub Size. Luer hub cavity is tested on the male luer taper gage of figure 5 of ASA Standard Z70-1-1955 (except with a suitable fluid path to permit performance of the fluid tightness test). The hub is slipped onto this gage with a light pressure. The fit of the hub on this gage is tested by means of the feeler gage of figure 5. The hub shall fit the plug gage within the "GO" "NO GO" limits of the feeler gage as shown in figure 5 and 6. In addition there shall be no movement of the hub on the plug gage when the cannula is pressed lightly from side to side. Hubs which do not fit the gages within these limits are not acceptable, except if plastic, a slight modification in taper is allowable provided it will insure a fluid and air tight fit between the hub and the gage.

4.3.2.8 Fluid and Air Tightness. The needle is tested in a horizontal position with the hub attached by means of

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the plug gage, described in 4.3.2.7, to water source having a pressure gage. Water is let into the needle to fill the hub cavity and lumen of the cannula. The cannula is sealed and a pressure of 45 pounds per square inch applied for 10 seconds. No leakage shall occur. The needle is attached to a syringe having a luer slip tip, no air leakage shall occur when water is aspirated into the syringe.

4.3.3 Safety (Toxicity) and Lead Tests.

4.3.3.1 Safety (Toxicity) Syringes and needles, as applicable, shall be tested for safety in accordance with the current edition of the U.S.P.

4.3.3.2 Test for Lead. Ten grams of rubber or plastic shall be taken, ignited to char, and then further ignited in a muffler at 700 degrees C. to a white ash. The ash shall be boiled in 15 Milliliters of 10 percent nitric acid filtered into a volumetric flask and made up to 100 Ml. An aliquot portion, equivalent to one gram of sample, shall be taken and placed in a separator. The dithizone test for lead, as given in the current edition of the U.S.P., shall be applied using a control of 5 p.p.m. or 12 p.p.m. C $\pm$ lead solution as given in the U.S.P. test. If the sample solution is darker than the limit, smaller aliquots shall be used until the color of the control is matched by the sample solution and the con-

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centration of lead in p.p.m. is determined. As an alternate 10 grams of rubber or plastic covered with 50 ml. of 0.9% saline solution and 10 grams covered with 50 ml. of cotton-seed oil (U.S.P.) warmed to 50°C. for 24 hours. A portion of each eluate shall be subjected to spectographic analysis for heavy metals.

TABLE VI

	STIFFNESS TEST			BREAKAGE TEST
GAGE	MAXIMUM DEFLECTION (INCH)	SEPARATION OF TEST EDGES (INCH) ± 0.003 INCH)	LOAD OUNCES (+ ½ oz.)	CONTACT DIST. FROM HUB (INCH)
27	0.022	0.375	20	0.3125
26	0.019	0.375	25	0.3125
25	0.026	0.500	25	0.4375
23	0.033	0.625	40	0.6250
22	0.029	0.625	50	0.6875
21	0.016	0.625	50	0.7500
20	0.021	0.750	50	1.0000
18	0.017	1.000	70	1.2500
19 T.W.	0.031.	1.000	70	1.1250

4.3.4 Freedom from Pyrogens. The pyrogen test shall be performed in accordance with the current edition of the U.S.P.

4.3.5 Sterility. The sterility tests shall be performed in accordance with the current edition of the U.S.P.

4.3.6 Freedom from Foreign Material. Syringes and needles, as specified, shall be tested as removed from individual containers. Style B syringes shall be tested with needle attached. The syringe and needles, as specified, shall be first examined with the naked eye for obvious foreign

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matter and shall then be tested as follows.

4.3.6.1 Test Material. Shall be high viscosity, soft, smooth textured white substance or paste. Before testing, the test material shall be examined closely under 36 power magnification for foreign coloring or matter. Any dark gray or black specks, spots or streaks shall indicate foreign particles and new test material selected and examined until suitable. A representative sample shall be retained for comparison and control purposes with the test specimen.

4.3.6.2 Test. A quantity of the test material sufficient to perform the test, but not less than ½ cubic centimeter shall be placed into each of five sample syringes with plungers removed. Plunger shall be carefully inserted and test material forced slowly through each syringe. Test specimens shall be first deposit forced through each syringe and spread on white bond paper. Presence of foreign matter in two or more syringes, determined by comparison with test material sample, both under 7 power magnification and under observation without magnification, using ordinary laboratory lighting, shall be cause for rejection.

4.3.7 Other tests for Type II Syringes. Type II syringes shall be tested for leakage of assembly, capacity, compression of bulb, and accuracy of graduations using water and a calibrated burette.

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5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Individual Container. Each syringe and needle, unless otherwise specified (see 6.1), shall be individually and securely packaged in a sealed, tamperproof container of appropriate size and design which shall maintain sterility until opened. The sealed container shall be so designed as to permit ready dispensing by sterile methods.

5.1.2 Unit of Issue and Packaging quantities shall be as specified (see 6.1).

5.1.2.1 Level A. Each unit of issue shall be packaged in a carton of appropriate size constructed in accordance with Federal Specification PPP-B-566, PPP-B-636, or PPP-B-676. Closure shall be adequate to prevent spillage of contents under normal handling. Commercial colors shall be acceptable on paperboard or corrugated cartons.

5.1.3 Packaging Quantities. The number of units of issue specified (see 6.1) shall be packaged in each unit, intermediate and exterior container, as applicable, for the required level of protection specified.

5.1.3.1 Packing Variation Permitted. If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable

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size and design, acceptable to a common carrier, which will insure safe delivery to destination.

5.1.4 Intermediate Package.

5.1.4.1 Level A. The number of units specified shall be packaged in a carton of appropriate size constructed in accordance with Federal Specification PPP-B-636, type CF, class domestic. Closure shall be adequate to prevent spillage of contents under normal handling.

5.1.4.2 Level C. The number of units specified shall be packaged in a standard commercial container of the size and kind commonly used for the purpose, which will afford the degree of protection required for shipment and use of the end product for its intended purpose within the continental limits of the United States of America.

5.2 Packing.

5.2.1 Level A.

5.2.1.1 Shipping Container. The number of units specified (see 6.1) shall be overpacked in an exterior container designed for type 2 load, and constructed in accordance with Federal Specification PPP-B-585, style 3, for class 3 use; PPP-B-601, table II, using type III, class 1, plywood; PPP-B-621, table III; or PPP-B-636, type CF, class weather resistant.

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5.2.1.2 Case Liner. Each level A wood box shall be lined with a waterproof case liner conforming to Military Specification MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiber board boxes.

5.2.1.3 Closure. Closure of wooden boxes shall be in accordance with the appendix of applicable box specification. Closure of each fiberboard box shall be as specified in the appendix, and waterproofing shall conform with 30.4 of PPP-B-636.

5.2.1.4 Strapping. Strapping, when required, shall be in accordance with appendix of applicable box specification.

NOTE: Strapping will not be required for shipments forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

5.2.2 Level B.

5.2.2.1 Shipping Containers. The number of units specified (see 6.1) shall be overpacked in an exterior container conforming to Federal Specification PPP-B-601, table I, using type III, class 1 plywood, or shall be designed for type 2 load, and constructed in accordance with Federal Specification PPP-B-585, style 3, for class 2 use; PPP-B-621, table I or PPP-B-636, type CF class domestic and method I closure shall be utilized.

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5.2.3 Level C. Federal Standard 102b, January 29, 1963. The number of units specified shall be packaged in accordance with suppliers commercial practice and packed to insure carrier acceptance and safe delivery in containers complying with rules and regulations applicable to the mode of transportation.

5.3 Marking. Markings shall be in contrasting colors and in such a manner as to be easily read.

5.3.1 Syringes. Each individual container and unit of issue package shall be marked or labeled with the applicable commercial description, (including capacity of syringe, gage and length of needle and bevel, if other than regular), name or registered trade mark of manufacturer, and legend indicating that the contents are sterile, pyrogen free, except type II syringe and disposable type. In addition the lot control number shall be clearly marked on each unit of issue container, or when unit of issue is each, on the intermediate package, and/or shipping container. Syringes and needles, as specified, that are incompatible with known parenteral drugs shall have a warning statement, prominently shown on or in the individual container and/or the unit of issue container, listing or defining the drugs that are not compatible with the materials used.

5.3.2 Needles. Each individual container and unit of issue package shall be marked or labeled with the applicable

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commercial description, including the gage, length of needle, and bevel, if other than regular. In addition the unit of issue container, or where the unit of issue is each on the intermediate package and/or shipping container, shall be marked or labeled with the applicable commercial description, name or registered trade mark or manufacturer, lot number and legend indicating that the contents are sterile, pyrogen free and disposable type.

5.3.3 Civilian Agencies. Shipping containers shall be marked in accordance with Federal Standard 123.

5.3.4 Military Agencies.

5.3.4.1 Unit Package. At least one side face of each unit or intermediate carton shall bear instructions for use and destruction as specified in 3.10. As an alternate printed instruction for use and destruction may be included in the unit or intermediate carton. The following shall also be shown:

Federal Stock Number

Item Identification

Quantity and Unit

Name (s) of manufacturer and Contractor

Lot (control) numbers

Legend indicating that syringes are sterile, pyrogen-free and disposable

Contract number

Level of preservation and date

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5.3.4.2 Intermediate Package. Each intermediate package shall be marked in accordance with Military Standard MIL-STD-129, except that when labels are utilized waterproofing shall not be required. Lot (control) number, contract or purchase order number and name of contractor shall be shown.

5.3.4.3 Shipping Container. Shipping container shall be marked in accordance with Military Standard MIL-STD-129. Lot (control) numbers shall be shown.

6. NOTES

6.1 Ordering data. Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

Title number and date of this specification

(a) Syringe, Type, Style, class and size (see 1.2)

(b) Needle, Gage, Length and point Style (if other than regular (see 1.2 and 3.2.3)

(c) Barrel Material (see 3.2.1.1)

(d) Hub material and Style (see 3.2.3.2)

(e) Design of Syringe Tip (see 3.2.1.1.1)

(f) Type of Tip (see 3.2.1.1.1)

(g) Graduation Scale (see 3.2.1.1.2.1) (Table I)

(h) Certificates (see 3.8)

(i) Selection of applicable levels of packaging and packing (see 5.1 and 5.2)

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(j) Bulk package (see 5.1.1)

(k) Unit of Issue (see 5.1.2)

(l) Packaging and Packing Quantities (see 5.1.3)

6.2 It is believed that this specification adequately describes the characteristics necessary to secure the desired material, and that normally no samples will be necessary prior to award to determine compliance with this specification. If, for any particular purpose, samples with bids are necessary, they should be specifically asked for in the invitation for bids, and the particular purpose to be served by the bid sample should be definitely stated, the specification to apply in all other respects.

6.3 Federal Specifications do not include all types, classes, grades, sizes, etc., of the commodities indicated by the title of the specification, or which are commercially available, but are intended to cover the types, etc., which are normally procured for Federal Government requirements.

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